Evaluation and Comparative Analysis

CSCI435/CSCI935 Computer Vision — Rice Leaf Disease Classification

1. The evaluation problem

The client needs a classifier that recognises Brown Spot, Leaf Scald, Rice Blast, Rice Tungro and Sheath Blight from either field photographs or images against a white background. A model that succeeds only on clean backgrounds may be unreliable in agricultural use. For that reason, results should be broken down by background as well as summarised across all images.

2. Methods represented in the group

Role 2 implements a conventional machine-learning baseline. It converts each image into colour and texture statistics and trains an RBF support vector machine. Its original code searches C values 0.1, 1, 10 and 100 and gamma values scale, 0.001, 0.01 and 0.1, selecting the combination with the highest validation macro F1. Role 5 implements a frozen ResNet-18 approach, using learned visual features from a pretrained network. The intended group comparison also includes a fine-tuned ResNet-18 from Role 6, but its trained predictions were not available in the files used for this evaluation.

3. How the preliminary experiment was run

The available Role 5 archive includes 1,106 thumbnails and five split records. The thumbnail baseline reused the HSV and LBP extraction from Role 2. To keep the local run manageable, the SVM used C = 1 and gamma = scale rather than searching the entire original parameter grid. Each split was trained independently, and test images were excluded from model fitting. The metrics and predictions are saved alongside this document for independent checking.

4. Summary of measured results

Background	Accuracy	Macro precision	Macro recall	Macro F1
White	48.95 ± 2.71	51.13 ± 4.14	45.24 ± 2.54	46.19 ± 2.72
Field	46.40 ± 7.13	48.12 ± 8.20	45.99 ± 7.51	45.17 ± 6.82
Combined	48.17 ± 2.93	48.47 ± 3.04	46.46 ± 3.49	46.66 ± 3.25

Values represent percentages, shown as the mean ± sample standard deviation over five runs. The combined-background accuracy of 48.17% should not be interpreted as a result for the original full-resolution SVM implementation or either ResNet variant.

5. What the results suggest

On average, white-background images produced slightly higher accuracy than field photographs. The difference was about 2.55 percentage points, while the variation between field-background runs was noticeably larger. This pattern is compatible with the idea that distracting visual context can make symptoms harder to isolate, although the present experiment does not establish a causal explanation.

The combined macro F1 of 46.66% is lower than combined accuracy. This is worth examining because macro F1 reflects the balance of precision and recall across the five diseases, whereas accuracy measures the fraction of correct predictions overall. A system intended for disease identification should avoid repeatedly missing one category even if its overall accuracy appears reasonable.

6. Confusion matrix and error analysis

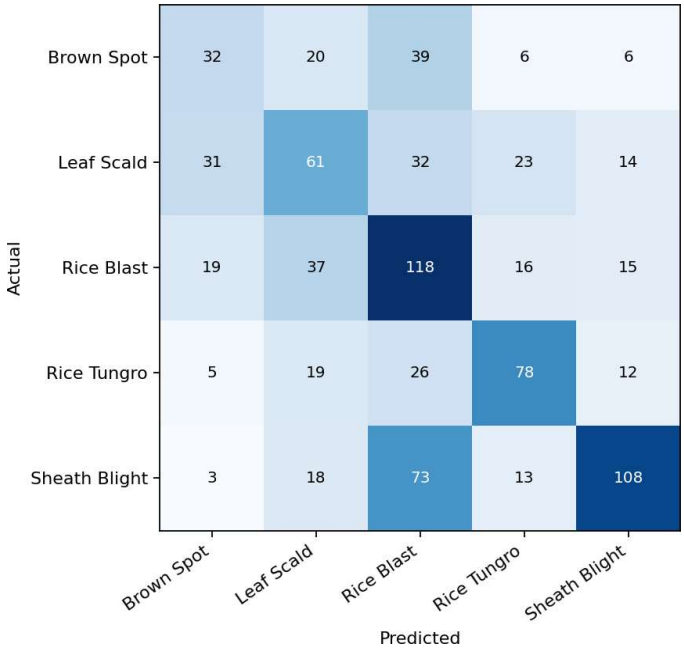


Figure 1. Pooled confusion matrix from the five thumbnail-based test runs. A sample can occur in multiple test runs and is therefore counted more than once.

The diagonal entries show correct classifications; off-diagonal entries indicate disease categories that the model confused. Because the same source image may appear in different runs, the pooled matrix is a diagnostic visualisation rather than a single independent test set. For more precise error analysis, the accompanying test_predictions.csv can be filtered by run, disease and background.

7. Fair comparison with the CNN approaches

A comparison between the SVM and ResNet approaches should hold the test samples constant. Otherwise, differences in performance could arise from the choice of images rather than from the model. The five seeds should identify the same train, validation and test images for every method. Training-time preprocessing may differ, but no test sample should be used to fit weights, select hyperparameters or train a scaler.

The frozen ResNet-18 approach is designed to retain pretrained convolutional features while learning a task-specific classifier. Fine-tuning would allow some pretrained features to adapt to rice leaf symptoms. Either approach could perform better or worse than the SVM depending on training settings, data quality and image characteristics; performance cannot be inferred reliably without test predictions.

8. Practical improvements

A stronger evaluation should include class-wise precision and recall, the complete five-run mean and standard deviation, and representative correctly and incorrectly classified images. Additional controlled experiments could test crop-based leaf isolation, consistent colour normalisation, and augmentations that simulate lighting changes. Any preprocessing fitted from data must be learned on the training subset alone.